Jaewoo Kim

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Korea Advanced Institute of Science and Technology (KAIST)
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RESEARCH INTERESTS

Space Systems Design & Operations

- Lifecycle-based space systems design considering stakeholder interactions and operational constraints
- Planning and management of space operations for efficient, resilient, and sustainable space utilization

Engineering Decision-Making Problems

- Formulation of complex engineering problems by identifying key system drivers
- Quantitative decision-making frameworks for interpretable and actionable engineering insights

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

Ph.D. in Aerospace Engineering *Feb. 2027 (Expected)*

- Thesis (in progress): Maintenance Strategies for Satellite Mega-Constellations (Advisor: Prof. Jaemyung Ahn)

M.S. in Aerospace Engineering *Feb. 2024*

- Thesis: Optimal Satellite System Architecting Considering On-Orbit Refueling (Advisor: Prof. Jaemyung Ahn)

Seoul National University (SNU) Seoul, Korea

B.S. in Aerospace Engineering *Feb. 2022*

- Thesis: Celestial Navigation Using Stars and Planets on Lunar Exploration Orbit (Advisor: Prof. Changdon Kee)

PUBLICATIONS

Journal Articles (Corresponding Authors Underlined)

[J1] Choi, E., **Kim, J.**, Kim, J.-H., Suh, E. S., and Ahn, J., "Optimal Design Structure Matrix Clustering with Constraint Selections for Conflict Resolution," accepted by *Journal of Mechanical Design*.

[J2] **Kim, J.**, Song, M., Lee, J., Yu, J., Choi, E. J., Kam, H., Jo, J., and Ahn, J., "Korean Space Collision Environment Assessment Framework Based on 3D-Cell Model," *International Journal of Aeronautical and Space Sciences*, 2026. doi:10.1007/s42405-026-01303-7

[J3] Park, B., Howell, K. C., **Kim, J.**, and Ahn, J., "Families of Two-Impulse Optimal Rendezvous Transfers Between Elliptic Orbits," *The Journal of the Astronautical Sciences*, Vol. 73, No. 5, 2026, pp. 75. doi:10.1007/s40295-026-00624-9

[J4] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "On-Orbit Servicing-Integrated Maintenance Strategy for Satellite Constellation," *Journal of Spacecraft and Rockets* (Online Published). doi:10.2514/1.A36684

[J5] **Kim, J.**, and Ahn, J., "System Architecting for GEO Communication Satellite Considering On-Orbit Refueling," *Journal of Spacecraft and Rockets*, Vol. 63, No. 1, 2026, pp. 176–191. doi:10.2514/1.A36403

[J6] **Kim, J.**, Choi, E., Ahn, J., Suh, E. S., Kim, J.-H., and Lim, D. G., "Mathematical Programming-Based Design Structure Matrix Clustering for Optimal Modular Architecture Design," *Journal of Mechanical Design*, Vol. 147, No. 10, 2025, pp. 101401.
doi:10.1115/1.4068100

[J7] **Kim, J.**, Ahn, J., and Sung, T., "Replenishment Strategy for Satellite Constellation with Dual Supply Modes," *Journal of Spacecraft and Rockets*, Vol. 62, No. 5, 2025, pp. 1576–1583.
doi:10.2514/1.A36281

[J8] Ko, J., **Kim, J.**, Choi, J., and Ahn, J., "Simultaneous Optimization of Launch Vehicle Stage and Trajectory Considering Flight-Requirement Constraints," *International Journal of Aeronautical and Space Sciences*, Vol. 25, 2024, pp. 1563–1573.
doi:10.1007/s42405-024-00737-1

[J9] Lee, D., **Kim, J.**, and Ahn, J., "Multi-Domain Matrix Framework for Human Resource Decision Support," under review at *Systems Engineering*.
doi:10.48550/arXiv.2607.01613

Papers Written in Korean

[K1] **Kim, J.**, Lee, J., Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "Development of 3D Cell Model for Collision Risk Assessment of Space Assets," *Journal of Space Technology and Applications*, Vol. 25, No. 2, 2025, pp. 114–124.
doi:10.52912/jsta.2025.5.2.114

Work in Progress

[P1] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "Joint Replenishment Strategy for Multiple Satellite Constellations with Shared Launch Opportunities," in preparation.
doi:10.48550/arXiv.2503.23666

[P2] **Kim, J.**, Choi, E., Suh, E. S., Kim, J.-H., and Ahn, J., "Managing Module Boundaries Across the OEM-Supplier Interface: A Bi-Objective DSM Clustering Approach," in preparation.

[P3] **Kim, J.**, Lee, S., Ahn, J., Choi, E., and Kim, J.-H., "Constrained Modularity Maximization in Complex Networks via Branch-and-Price," in preparation.

[P4] **Kim, J.**, Ahn, J., and Choi, J., "Reinforcement Learning-Based Architecture Design Optimization of Large Space Systems with On-Orbit Servicing," in preparation.

[P5] **Kim, J.**, Song, M., Yu, J., Choi, E. J., Kam, H., Jo, J., and Ahn, J., "Cooperative Tasking of Space and Ground Observatories for Emergency Collision Events," in preparation.

Proceedings of Refereed Conferences

[C1] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "Impact and Mitigation of Launch Failures on Satellite Constellation Maintenance," in *AIAA SciTech Forum*, Orlando, Florida, US, Jan. 11–15, 2027. (to appear)

[C2] Lee, S., **Kim, J.**, and Ahn, J., "A Multi-Trip Time-Dependent Traveling Salesman Problem for Space Logistics," in *2026 INFORMS Annual Meeting*, San Francisco, California, US, Nov. 1–4, 2026. (to appear)

[C3] **Kim, J.**, Song, M., Lee, J., Yu, J., Kam, H., Choi, E.-J., Choi, J., Jo, J. H., and Ahn, J., "Inclination-Aware 2D Source-Sink Modeling for Long-Term Projection of the LEO Collision Environment," in *27th Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii, US, Sep. 15–18, 2026.
[.pdf Link]

[C4] Song, M., **Kim, J.**, and Ahn, J., "Decision Support for Space Debris Remediation Enterprises via Two-Stage System Optimization under Various Future Scenarios," in *27th Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii, US, Sep. 15–18, 2026.

[C5] **Kim, J.**, Choi, J., and Ahn, J., "Flexibility-Aware Assessment and Optimization of System Architectural Decision for Large Space Systems Considering On-Orbit Servicing," in *Asian Congress of Structural and Multidisciplinary Optimization 2026 (ACSMO 2026)*, Busan, Korea, May 17–21, 2026.

[C6] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "Integration of On-Orbit Servicing into Constellation Maintenance with Spare Inventory Management," in *AIAA SciTech Forum*, Orlando, Florida, US, Jan. 12–16, 2026. doi:10.2514/6.2026-2631

[C7] Song, M., **Kim, J.**, Nam, Y., and Ahn, J., "Low Earth Orbit Space Debris Remediation Program for Maximizing Stakeholders' Utility," in *Asia-Pacific International Symposium on Aerospace Technology (APISAT) 2025*, Seoul, Korea, Oct. 27–29, 2025.

[C8] **Kim, J.**, Sung, T., and Ahn, J., "Joint Replenishment Strategy for Multiple Satellite Constellations," in *AIAA SciTech Forum*, Orlando, Florida, US, Jan. 6–10, 2025. doi:10.2514/6.2025-1757

[C9] **Kim, J.**, Lee, J., Kim, H., Choi, E. J., Choi, J., Yu, J., Jo, J., and Ahn, J., "Development of Korea Orbital Debris Evolutionary and Engineering Model," in *75th International Astronautical Congress*, Milan, Italy, Oct. 14–18, 2024. doi:10.52202/078360-0161

Proceedings Written in Korean

[C10] **Kim, J.**, Song, M., Yu, J., Kam, H., Jo, J. H., Choi, E. J., Choi, J., and Ahn, J., "Progress in Korean Space Collision Environment Models," in *Korean Space Science Society (KSSS) Fall Conference*, Seoul, Korea, Nov. 5–6, 2026. (to appear)

[C11] Yi, G., **Kim, J.**, and Ahn, J., "A Review of Modeling and Optimization Approaches for Time-Dependent Space Logistics Network," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Jeongseon, Korea, Nov. 10–13, 2026. (to appear)

[C12] **Kim, J.**, Song, M., Ahn, J., Yu, J., Kam, H., Jo, J. H., Choi, J., and Choi, E.-J., "Development of Integrated Framework for Collision Environment Assessment," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Jeongseon, Korea, Nov. 10–13, 2026. (to appear)

[C13] Lee, S., **Kim, J.**, Ahn, J., and Kim, J.-H., "Constrained Modularity Maximization in Complex Networks," in *Korean Society for Industrial and Applied Mathematics (KSIAM) Fall Conference*, Jeju, Korea, Nov. 5–8, 2026. (to appear)

[C14] Song, M., Lee, J., **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "3D-Cell Model Analysis and Application for Korean Space Collision Environment Assessment Framework," in *Korean Space Science Society (KSSS) Fall Conference*, Jeju, Korea, Oct. 27–30, 2026. (to appear)

[C15] **Kim, J.**, Song, M., Choi, E.-J., Choi, J., Yu, J., Jo, J. H., Kam, H., and Ahn, J., "2D Evolutionary Model for Assessment of Future LEO Collision Environment," in *Korean Space Science Society (KSSS) Fall Conference*, Jeju, Korea, Oct. 27–30, 2026. (to appear)

[C16] **Kim, J.**, Song, M., Choi, E.-J., Choi, J., Yu, J., Jo, J. H., Kam, H., and Ahn, J., "Optimal Space-Ground Sensor Tasking for Cooperative Observation of Space Objects," in *Korean Space Science Society (KSSS) Fall Conference*, Jeju, Korea, Oct. 27–30, 2026 (to appear).

[C17] **Kim, J.**, Ahn, J., Park, B., and Howell, K. C., "A Family-Based Framework for Two-Impulse Optimal Rendezvous Transfers," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Busan, Korea, Jun. 24–26, 2026.

[C18] **Kim, J.**, Song, M., Yu, J., Kam, H., Choi, E. J., Choi, J., Jo, J. H., and Ahn, J., "3D Cell Model-Based Collision Risk Analysis for Historical and Prospective LEO Scenarios," in *Korean Academy of Space Security (KASS) Spring Conference*, Yeosu, Korea, May 14–15, 2026.

[C19] **Kim, J.**, Song, M., Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "Updates on Space Collision Risk Analysis Framework," in *Korean Space Science Society (KSSS) Spring Conference*, Yeosu, Korea, Apr. 22–24, 2026.

[C20] **Kim, J.**, Yi, G., and Ahn, J., "Implications of In-situ Resource Utilization from Space Logistics Perspective," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Mar. 31–Apr. 3, 2026.

[C21] **Kim, J.**, Yi, G., and Ahn, J., "Research and Standardization of Space Logistics for Long-term Complex Space Mission," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Mar. 31–Apr. 3, 2026.

[C22] **Kim, J.**, Sung, T., Song, M., and Ahn, J., "Integrated Inventory Model for Constellation Maintenance via Spare Satellites and On-Orbit Servicing," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Goseong, Korea, Nov. 11–14, 2025.

[C23] **Kim, J.**, Song, M., and Ahn, J., "Optimal Design Lifetime and Propellant Loading of GEO Communication Satellites Considering On-Orbit Refueling," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Goseong, Korea, Nov. 11–14, 2025.

[C24] **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "Survey of Risk Indices for Space Objects and Implications," in *Korean Space Science Society (KSSS) Fall Conference*, Jeju, Korea, Oct. 29–Nov. 1, 2025.

[C25] Lee, J., Lee, J., Jang, E., Jeon, H., **Kim, J.**, Lee, C., Scheeres, D. J., and Han, J., "Asteroid Accessibility Searching Algorithm," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Yeosu, Korea, Jun. 25–27, 2025.

[C26] **Kim, J.**, Song, M., and Ahn, J., "Survey of Research on Economic and Performance Analysis of On-Orbit Servicing and Implications," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Yeosu, Korea, Jun. 25–27, 2025.

[C27] **Kim, J.**, Ahn, J., Choi, E. J., Choi, J., Yu, J., and Jo, J., "Development of Integrated Space Collision Risk Response System," in *Korean Academy of Space Security (KASS) Spring Conference*, Seoul, Korea, May 9, 2025.

[C28] **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., and Ahn, J., "Progress of Space Debris Environment and Risk Analysis Framework Development," in *Korean Space Science Society (KSSS) Spring Conference*, Seoul, Korea, Apr. 22–24, 2025.

[C29] **Kim, J.**, Sung, T., and Ahn, J., "Application of Space Logistics: Maintenance Strategy of Satellite Mega-Constellation," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 2–4, 2025.

[C30] **Kim, J.**, Lee, J., Choi, E. J., Jin, C., Yu, J., Jo, J., and Ahn, J., "Development of Korean 3D Cell Model for Space Debris Environment Analysis," in *Korean Space Science Society (KSSS) Fall Conference*, Sacheon, Korea, Oct. 28–30, 2024.

[C31] **Kim, J.**, and Ahn, J., "An Integrated Inventory Management Model for Maintenance of Multiple Satellite Constellations," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Changwon, Korea, Jun. 26–28, 2024.

[C32] **Kim, J.**, Ko, J., Choi, J., Ahn, J., Yoon, N., and Kim, H., "Conceptual Design of Launch Vehicle Considering Axial Acceleration Constraints," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Changwon, Korea, Jun. 26–28, 2024.

[C33] Ko, J., **Kim, J.**, Choi, J., Ahn, J., Yoon, N., and Kim, H., "Development of Conceptual Design Software for Space Launch Vehicle," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 3–5, 2024.

[C34] **Kim, J.**, and Ahn, J., "Multiobjective Design Optimization of Commercial Satellite Considering On-Orbit Refueling Policy," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 19–21, 2023.

[C35] **Kim, J.**, Lee, D. U., and Ahn, J., "Research on the Overseas On-Orbit Servicing Trends and Implications," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Jeju, Korea, Nov. 16–18, 2022.

RESEARCH EXPERIENCE

Graduate Research Assistant | *Strategic Aerospace Initiative, KAIST* *Mar. 2022 – Present*

- **In-Space Servicing and Manufacturing Research Center (ISMRC)**
  Duration: 2025–2035; Sponsor: National Research Foundation of Korea
  - Developed a maintenance framework for satellite mega-constellations integrating on-orbit servicing [C6]
  - Surveyed standardization of space logistics framework [C20, C21]
- **Space Logistics Modeling and Demand Fulfillment Strategy Evaluation Framework**
  Duration: 2025–2028; Sponsor: National Research Foundation of Korea
  - Developed frameworks for evaluating the value of on-orbit servicing and potential architectural changes for GEO communication satellites [J5, C5, C23, C26]
  - Proposed maintenance strategies for satellite mega-constellations [J4, P1, C1, C6, C8, C22, C29]
  - Supported conceptual design for a space debris remediation program [C4, C7]
  - Surveyed in-situ resource utilization from a logistics perspective [C20]
- **Multi-Criteria Decision Making for Optimal Modular System Architecture Selection**
  Duration: 2024–2026; Sponsor: Hyundai Motor Company
  - Surveyed multi-criteria decision-making methods and their applications in system architecting
  - Developed a decision-making framework for design structure matrix-based modular architecture design considering OEM-supplier interfaces [P2]
  - Supported the development of a mathematical programming approach for modular architecting with conflict resolution among design constraints [J1]
  - Developed a mathematical programming-based framework for constrained modularity maximization [P3, C13]
- **Development of an Integrated Space Risk Response System**
  Duration: 2024–2027; Sponsor: Korea Astronomy and Space Science Institute
  - Surveyed models for the space object environment and collision-risk analysis
  - Developed engineering and evolutionary models of space collision environment for Korean Integrated Space Risk Response System [J2, K1, C3, C9, C10, C12, C14, C15, C18, C19, C24, C27, C28, C30]
  - Developed a ground-space cooperative sensor tasking framework for operational planning [P5, C16]
- **A Study on the Principle of Modular Architecture Engineering to Improve Level of Completion for Vehicle Architecture**
  Duration: 2022–2023; Sponsor: Hyundai Motor Company
  - Developed a mathematical programming approach for design structure matrix-based modular architecting [J6]
  - Conducted case studies on automobile subsystems and identified improved design solutions
- **Research on ADR/OOS Applications for National Security Space Assets**
  Duration: 2022; Sponsor: Korean Society for Aeronautical and Space Sciences
  - Reviewed on-orbit servicing technologies and related mission concepts [C35]
  - Developed concepts of operations for ADR/OOS missions
- **Development of Launch Vehicle Mission & Conceptual Design Software**
  Duration: 2022–2023; Sponsor: Hanwha Aerospace
  - Developed propulsion and staging modules for multidisciplinary design optimization
  - Contributed to the development of an all-at-once design optimization framework considering flight requirements [J8, C32, C33]

Undergraduate Researcher | *GNSS Laboratory, SNU* *Mar. 2021 – Aug. 2021*

- **Deep Space Navigation with Optical Sensor Data**
  - Reviewed non-inertial deep-space navigation algorithms
  - Evaluated the selected algorithm through numerical simulations

Teaching Experience

Teaching Assistant | *KAIST* *Sep. 2023 – Present*

- Courses: AE210 Aerospace Thermodynamics ('24 S), AE280 Software Application in Aerospace Engineering ('26 S, '25 S), AE401 Aerospace System Design II ('23 F), AE450 Flight Dynamics and Control ('24 F), AE551 Introduction to Optimal Control ('26 F, '25 F)

Part-Time Lecturer | *Humaiin Co.* *Aug. 2022 – Present*

- Delivered lectures and developed instructional materials covering foundational data science concepts, programming tools, and LLM-based workflow automation.
- Institutions: Busan City Government, Korea Education & Research Information Service (KERIS), Statistics Korea (KOSTAT), Ewha Womans University, Sookmyung Women's University, Seoul AI Foundation, Hanwha Solutions, Lotte Chemical

Professional Services

- Reviewer for *Systems Engineering*, *International Journal of Aeronautical and Space Sciences*; *AIAA SciTech Forum*, *AIAA AVIATION Forum*.

Awards & Honors

2025 Korean Space Science Society Spring Conference | *Best Oral Presentation Award* *Apr. 2025*

- Presentation Title: Progress of space debris environment and risk analysis framework development

Optimization Grand Challenge 2024 | *Top 10% of Participants* *Oct. 2024*

- Competition to develop an optimization algorithm for pickup-and-delivery problems, sponsored by LG CNS
- Team Name: J2Opt

Hanwha-KAIST Space Hub Space Grand Challenge | *Bronze* *Nov. 2023*

- Competition for innovative space systems, technologies, and mission concepts, sponsored by Hanwha Aerospace
- Team Name: LETA (Lunar Exploration Trajectory Analytics)
- Topic: Lunar exploration trajectory design with low-thrust propulsion and multiple gravity assist
- Prize: 3,000,000 KRW (2,050 USD)

Extracurricular Experience

Military Service | *Defense Security Command (DSC)* *Apr. 2018 – Nov. 2019*

- Supported educational programs for military officers in DSC, and served as a squad leader
- Received commendation from DSC School Head (Brigadier General)

Interviewer | *Humans of SNU* *Jul. 2017 – Dec. 2017*

- Interviewed diverse members of the SNU community and discovered insightful and interesting stories

Volunteer | *People to People International* *Mar. 2016 – Feb. 2018*

- Supported underprivileged communities in Seoul
- Student chapter union president (2017); SNU chapter president (2016)

OTHER SKILLS

Programming

- Python, MATLAB, Julia, C, C++, Java for quantitative analysis, including optimization, simulation, and statistical reasoning

Languages

- Korean (native), English